
بسم الله الرحمن الرحيم

Electromagnetic Radiations, Spectrum and Energy Interaction

Introduction

- Interpretation of remote sensing imagery depends on a sound understanding of electromagnetic radiation and its interaction with the atmosphere and the surfaces
- Remote sensing relies on the measurement of electromagnetic (EM) energy
- EM energy can take several different forms
- The most important source of EM energy at the Earth's surface is the Sun

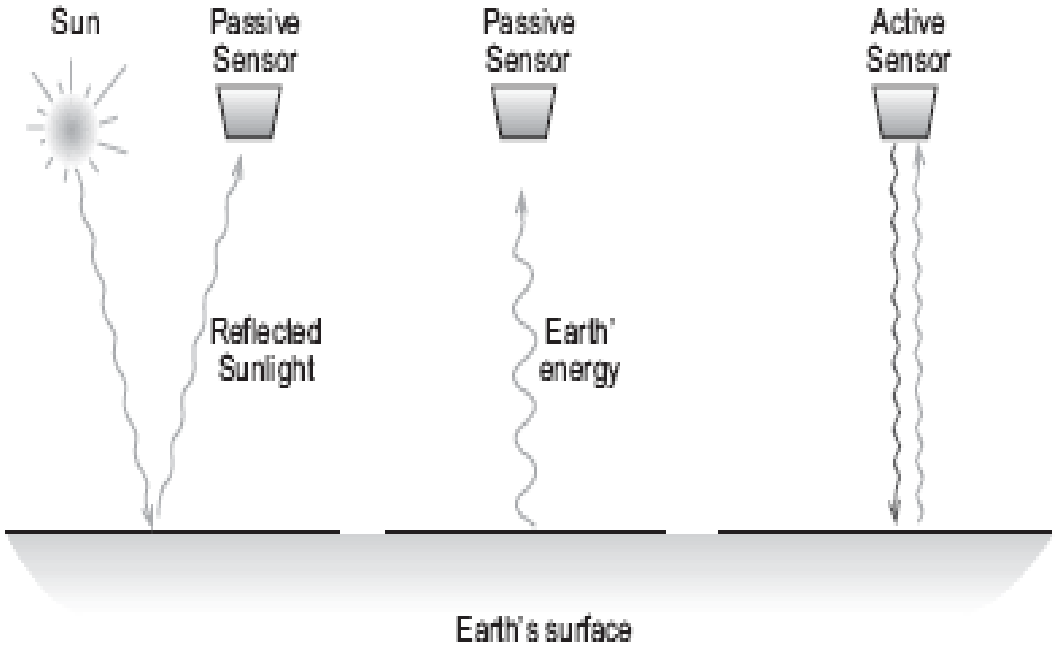
Sensors

- Many sensors used in remote sensing

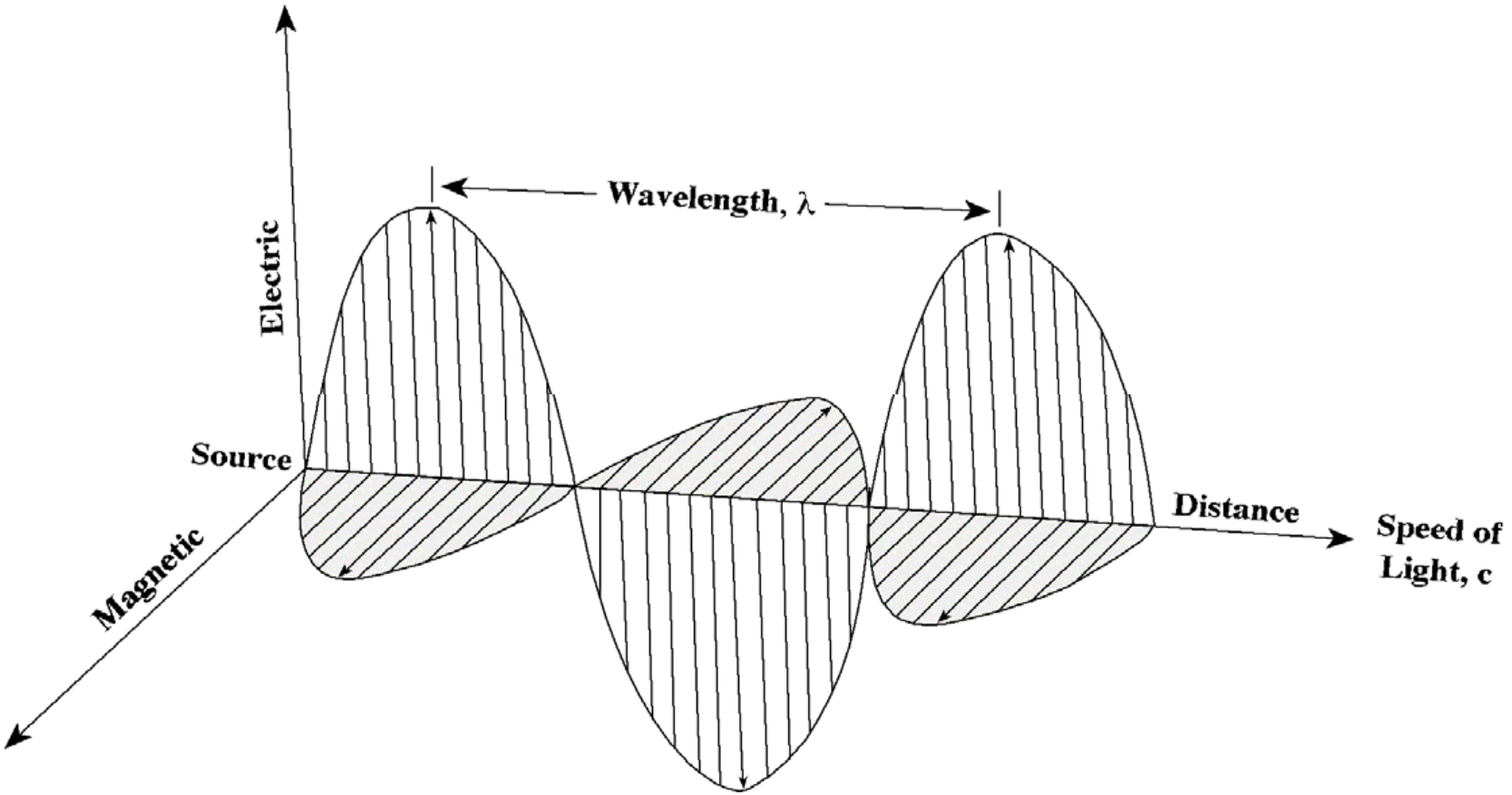
  measure reflected sunlight.
- Some sensors, however, detect energy

  emitted by the Earth itself or provide their

  own energy.



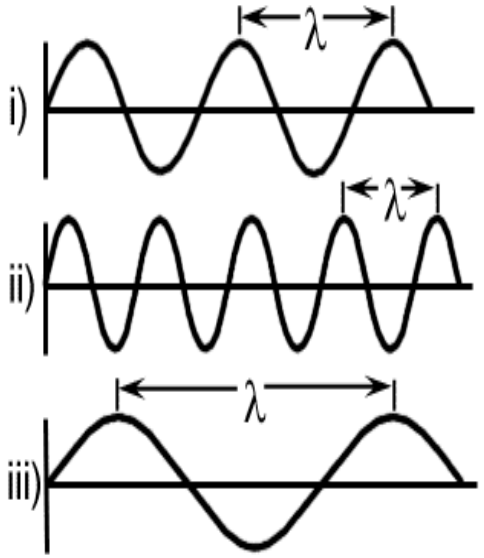
Sinusoidal Wave Propagating Through Space and Time



Electric field is perpendicular to the magnetic field

Important terms

- Wavelength: The distance between from one wave crest to the next
- Frequency: The number of crests passing a fixed point in a given period of time
- Amplitude: Height of each peak, measured as energy level per wavelength interval
- Relationship between wavelength and frequency
  - Wavelength (μm): $\lambda = c / \nu$
  - Frequency (Hz): $\nu = c / \lambda$



How EM energy interact with matter

- Rate at which Photons strike a surface is termed as *Radiant Flux* (Φ_e)*measured in Watts (W)*
- *Radiant flux* per unit area is termed as *Irradiance*(E_e), *usually* measured as watts per square meter. *Irradiance* measures radiations that strikes a surface
- The rate at which radiation is emitted from a unit area is termed as Radiant exitance (M_e) also measured as watts per square meter
- All objects with temperature above zero emit energy
- Amount of energy and the wavelength at which it is emitted depends upon the temperature of object
- As temperature increases, the amount of emitted energy also increases and wavelength of maximum emission becomes shorter

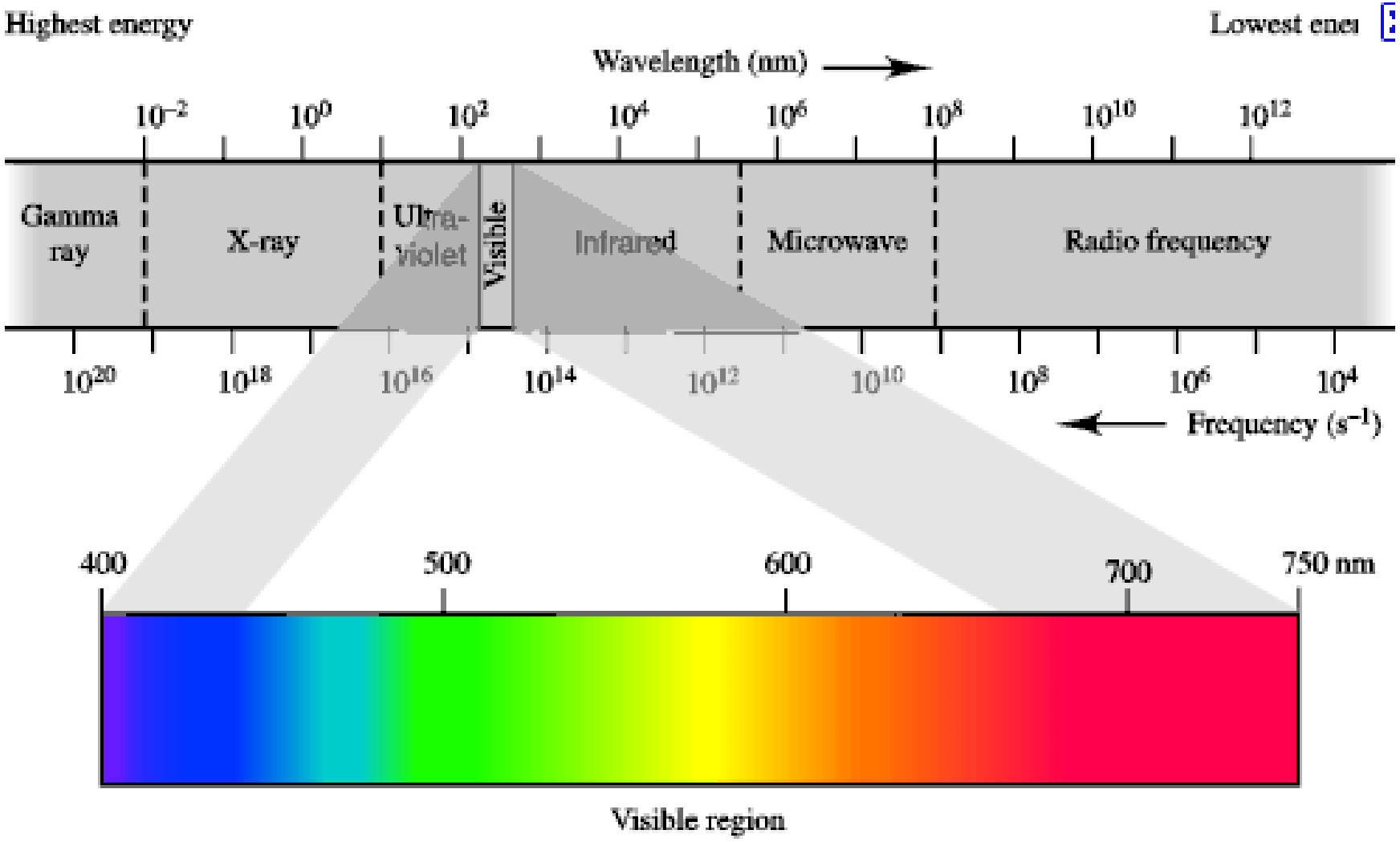
Symbol	Description	Defining Expression	Units	Units (Abbreviation)
Q	Radiant energy		Joules	J
Φ	Radiant flux (power)	$\frac{dQ}{dt}$	Watts	W
E	Irradiance	$\frac{d\Phi}{dA}(in)$	Watts per square meter	W/m^2
M	Radiant Exitance	$\frac{d\Phi}{dA}(out)$	Watts per square meter	W/m^2
L	Radiance	$\frac{d^2\Phi}{\cos\theta\, dA\, d\omega}(out)$	Watts per square meter per steradian	$\frac{W}{(m^2 - sr)}$
E_λ	Spectral Irradiance	$\frac{dE}{d\lambda}(in)$	Watts per square meter per micrometer	$\frac{W}{(m^2 - \mu m)}$
M_λ	Spectral Radiant Exitance	$\frac{dM}{d\lambda}(out)$	Watts per square meter per micrometer	$\frac{W}{(m^2 - \mu m)}$
L_λ	Spectral Radiance	$\frac{dL}{d\lambda}$	Watts per square meter per steradian per micrometer	$\frac{W}{(m^2 - sr - \mu m)}$

Table 2-1. A radiometric system of units for remote sensing quantities.

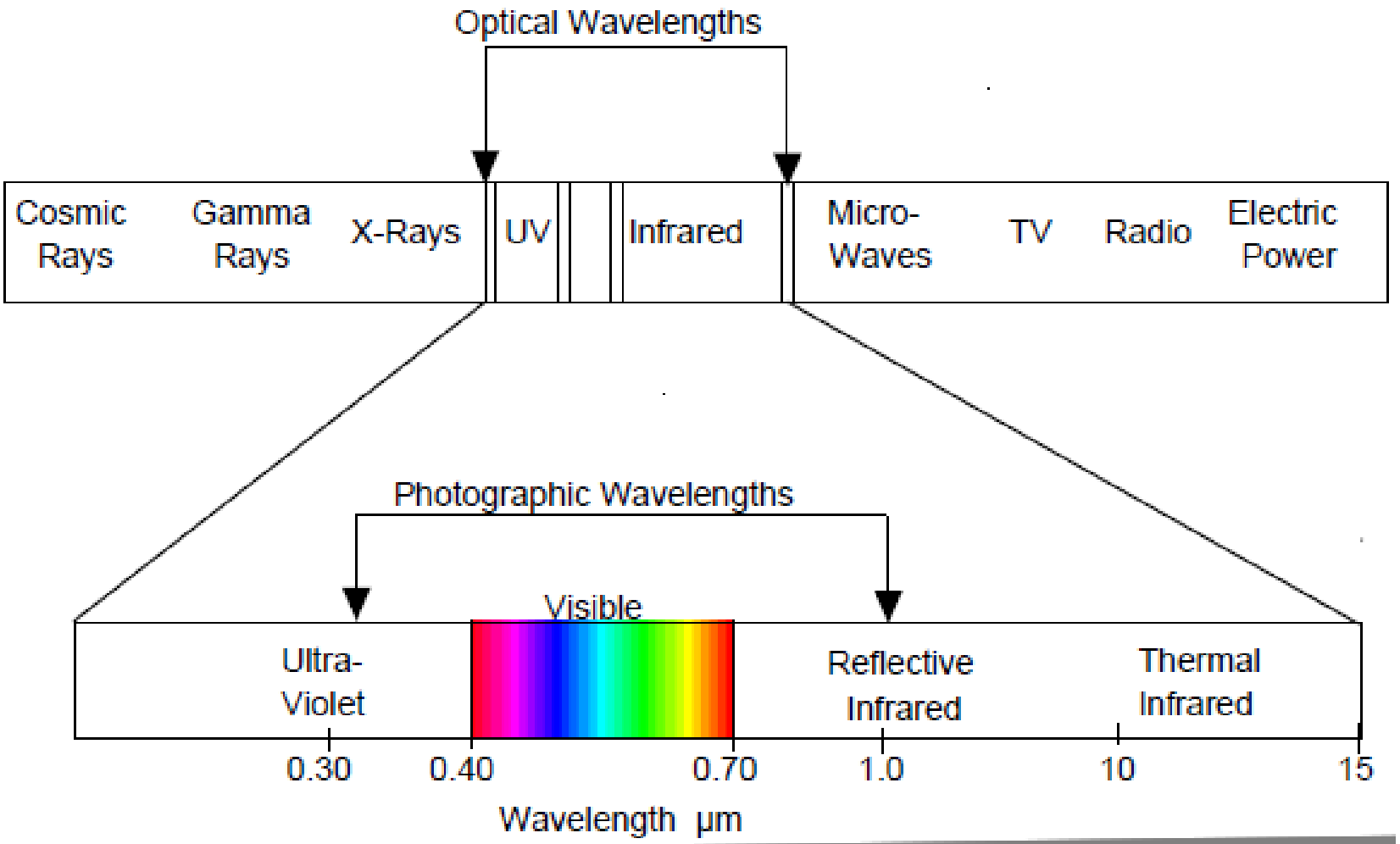
Electromagnetic Spectrum

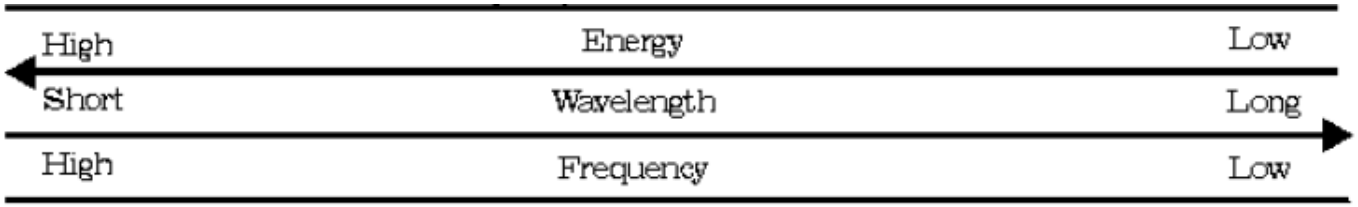
- Electromagnetic Spectrum ranges from the shorter wavelengths (including gamma and x-rays) to the longer wavelengths (including microwaves and broadcast radio waves)
- There are several regions of the electromagnetic spectrum which are useful for remote sensing

EMR SPECTRUM



EM spectrum is arbitrarily segmented into major divisions. There are no natural breaks in the EM spectrum. These separations are made by us for our convenience.





Electromagnetic Spectrum

- Light which our eyes- and “remote sensors” – can detect is part of the *Visible Spectrum*
- Visible portion is much small relative to the rest of the spectrum
- A lot of radiations around us are “invisible” to our eyes, but can be detected by other remote sensing instruments and used to our advantage

Electromagnetic Spectrum

- *Infrared (IR)* region covers the wavelength range from approximately 0.7 μm to 15 μm – more than 40 times as wide as the visible portion
- Infrared region can be divided into two categories based on their radiation properties
  - **Reflected IR,** (Near IR)- essentially solar radiation reflected from Earth's surface **(**0.7 μm to 3.0 μm)
  - **Emitted or Thermal** IR (Far IR)- radiations emitted by the Earth, consists of Heat and thermal energy (3.0 μm to 14.0 μm)

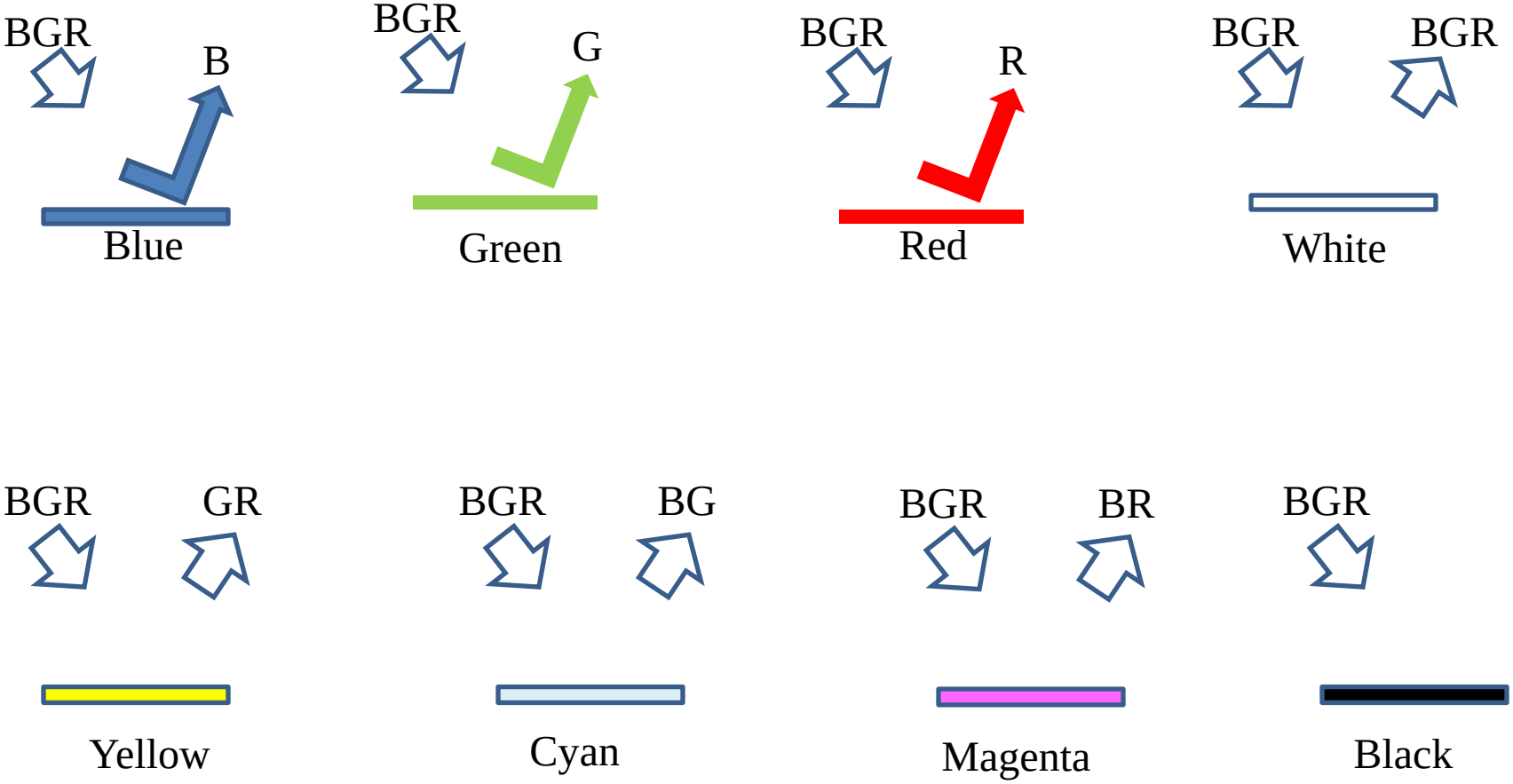
Electromagnetic Spectrum

- Radiation in the reflected IR region is used for remote sensing purposes in ways very similar to radiation in the visible portion
- Thermal IR region is quite different than the visible and reflected IR portions, as this energy is essentially the radiation that is emitted from the Earth's surface in the form of heat

Division Of Electromagnetic Spectrum

Division	Range
Gamma rays	< 0.03 nm
X-rays	0.03 – 300 nm
UV radiations	0.30 – 0.38 μm
Visible light	0.38 -.72 μm
Infrared radiations (Near IR)	0.72- 1.3 μm
Mid IR	1.3- 3.00 μm
Far IR	7.0 μm – 1
Microwaves radiations	1 mm- 30 cm
Radio	> 30 cm

The color of an object is defined by the color of the light it reflects



Interaction with the Atmosphere

- Atmospheric modification of incoming and outgoing EM radiation includes:
  - Scattering: direction unpredictable
  - Refraction: the direction predictable
  - Absorption

Interactions With The Atmosphere

Scattering

- Occurs when particles or large gas molecules present in the atmosphere interact with and cause the electromagnetic radiation to be redirected from its original path
- Scattering depends on several factors including
  - Wavelength of the radiation
  - Size and abundance of suspended particles or gases
  - Distance the radiation travels through the atmosphere

Interactions With The Atmosphere

- ***Rayleigh Scattering:*** **dominant scattering mechanism in the upper atmosphere, also called clear atmosphere scattering**
- **Occurs when particles are very small (dust or nitrogen / oxygen molecules) as compared to the wavelength of the radiation**
- **Is wavelength dependent, scattering increases as wavelength becomes shorter**
- **Causes shorter wavelengths of energy to be scattered much more than longer wavelengths**

Interactions With The Atmosphere

- The fact that the sky appears “blue” during the day is because of Rayleigh Scattering.
- As sunlight passes through the atmosphere, the shorter wavelengths (i.e. blue) of the visible spectrum are scattered more than the other (longer) visible wavelengths

Interactions With The Atmosphere

- At sunrise and sunset the light has to travel farther through atmosphere than at midday and the scattering of the shorter wavelengths is more complete; this leaves a greater proportion of the longer wavelengths to penetrate the atmosphere

Interactions With The Atmosphere

Mie Scattering

- Occurs when the particles are just about the same size as the wavelength of the radiation (Dust, pollen, smoke and water vapour)
- Tends to affect longer wavelengths
- Occurs mostly in the lower portions of the atmosphere(0-5 km) where large particles are more abundant

Interactions With The Atmosphere

Nonselective Scattering

- All wavelengths are scattered about equally
- Occurs when the particles are much larger than the wavelength of the radiation (Water droplets and large dust particles)
- Causes fog, haze and clouds to appear white to our eyes because blue, green, and red light are all scattered in approximately equally

General Effects Of Scattering

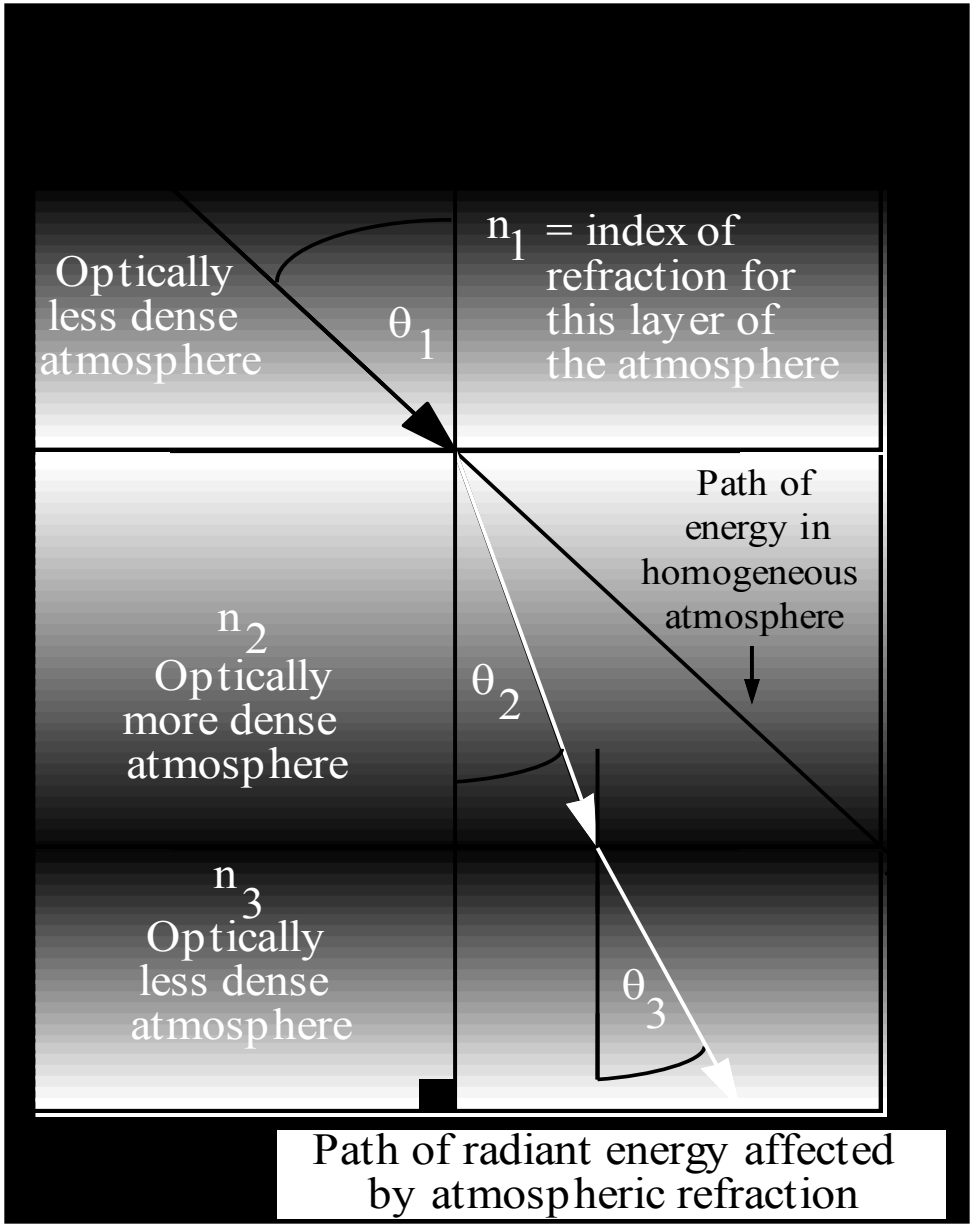
- Causes skylight scattering which allows us to see items within a shadow
- Forces image to record the brightness of the atmosphere in addition to the target
- Reduces contrast

Atmospheric Refraction

- Speed of light c is 3×10^8 m s^{-1} (same as Electromagnetic Radiation EMR)
- When encounters substances of different density (air and water), refraction may take place
- Refraction: bending of light when it passes from one medium to another
  - Refraction occurs because the media are of differing densities and the speed of EMR is different in each
- The index of refraction, n: measure of the optical density of a substance
  - This index is the ratio of c, to the speed of light in the substance, c_n:

$$n = \frac{\text{speed of light in vacuum}}{\text{speed of light in the medium}}$$

Atmospheric Refraction



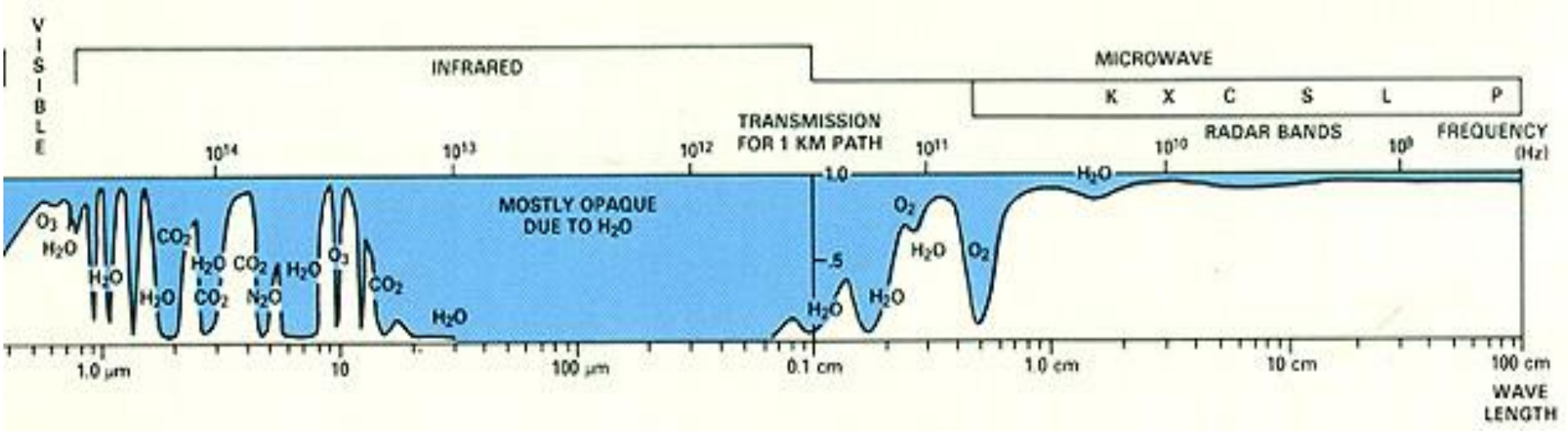
Interactions with the Atmosphere

Absorption

- **This phenomenon causes molecules in the atmosphere to absorb energy at various wavelengths.**
- **Ozone, carbon dioxide and water vapour are the three main atmospheric constituents which absorb radiation**
- **Because these gases absorb electromagnetic energy in very specific regions of the spectrum, they influence where (in the spectrum) we can "look" for remote sensing purpose**

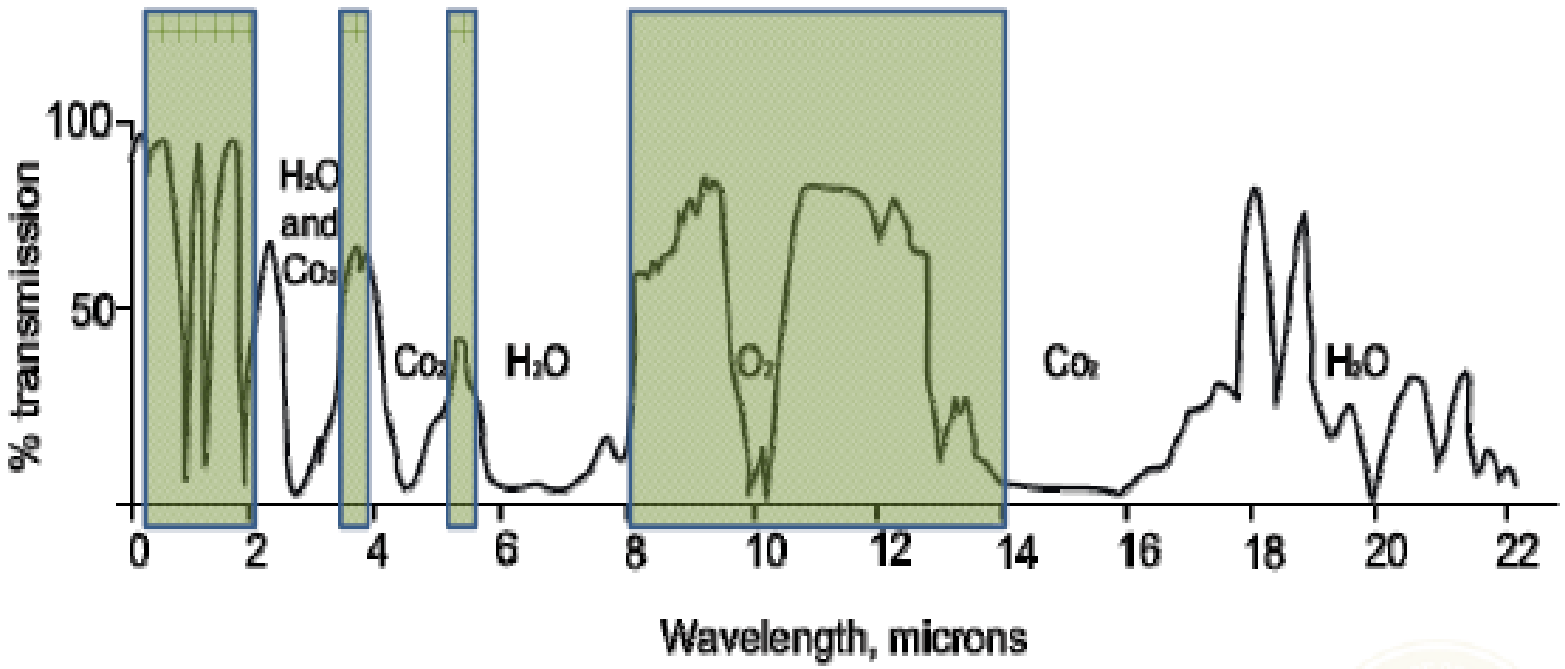
Atmospheric Transmission

Generalized diagram showing relative atmospheric transmission of radiation of different wavelengths.



Blue zones mark minimal passage of incoming and/or outgoing radiation whereas white areas denote "atmospheric windows" in which the radiation experiences much reduced interactions and hence can penetrate the air with little or no loss by absorption.

Atmospheric Windows



- Window between 0.4-2 μm where the (optical) remote sensors operate as well as the human eye
- Windows in the thermal infrared region, around 3 and 5 μm,
- Relatively broad, window extending from approximately 8 to 14 μm

Interaction with Earth Features

- Radiations that are not absorbed or scattered in the atmosphere reaches and interact with the Earth's surface
- Interaction with the target depends upon the properties of both the target and the radiation
- Three things happen to the sun's energy as it reaches the Earth:
  - Reflected
  - Absorbed
  - Transmitted
- The proportions of each will depend on the wavelength of the energy, the material and condition of the feature

Reflection

- When a surface is smooth, we get specular or mirror - like reflection where all (or almost all) of the energy is directed away from the surface in a single direction
- Specular reflection can be caused, for example, by a water surface or a glasshouse roof
- It results in a very bright spot in the image

Reflection

- **Diffuse reflection occurs when the surface is rough and the energy is reflected almost uniformly in all directions**

- **Whether a particular target reflects specular or diffusely, or somewhere in between, depends on the surface roughness of the feature in comparison to the wavelength of the incoming radiation**

Recording Of Energy By The Sensor

- **After the energy has been reflected, scattered by, or emitted from the target, a sensor collects and records the electromagnetic radiation**

Reference/Reading Assignment

- Chapter 2, RS of Environment, J R. Jensen
- Sections 1.2, 1.3, Lillesand
- Chapter 1, Sections 1.2, P M Mather
- Chapter 2, J B. Campbell

- Aerial Photography and Image Interpretation, book